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FINAL REPORT

RECEIPTS OCR AND INFORMATION EXTRACTION

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Abstract

This project proposes an automated approach for extracting key information from receipt images using modern document understanding techniques. Processing receipts is challenging due to their semi-structured layouts, varying formats and complex visual structures. To address these challenges, the proposed system combines optical character recognition (OCR) with a multimodal document understanding model to accurately identify important information.

In the first stage, PaddleOCR is used to detect text regions and recognize textual content from receipt images. The extracted text is then associated with its corresponding spatial layout information, including the bounding box coordinates of each text segment. In the second stage, the LayoutLMv3 model is applied to capture relationships between textual, layout and visual features for document understanding and key information extraction.

The system is trained and evaluated on the SROIE2019, which is a benchmark dataset for receipt information extraction tasks. Experimental results show that integrating OCR techniques with a multimodal document model improves the accuracy of extracting structured information such as company name, address, date and total amount. The proposed approach demonstrates the potential of deep learning–based document AI techniques in automating receipt processing for applications in accounting and financial management.

1. Introduction

Many organizations process large numbers of receipts for financial management and expense tracking. These documents are often stored as scanned images or photographs, making manual data entry time-consuming and prone to human errors. Therefore, developing automated methods for extracting information from receipt images has become increasingly important.

Receipt information extraction aims to automatically identify and extract key fields such as company name, date, address and total amount. However, this task is challenging due to variations in receipt layouts, different image qualities and multiple text regions distributed across the document.

In this project, we develop a receipt information extraction pipeline and evaluate its performance.. The goal of this work is to explore how computer vision techniques can be applied to automatically extract structured information from receipt images.

2. Dataset

2.1 Overview of SROIE2019

The SROIE2019 (Scanned Receipts OCR and Information Extraction) dataset was introduced in the ICDAR 2019 Robust Reading Challenge to support research on automatic receipt understanding. The dataset focuses on extracting key information from scanned or photographed receipts, which is a common task in document intelligence systems.

The dataset contains receipt images collected from various stores with different layouts, fonts and visual qualities. Each receipt image is paired with annotations that specify important entities such as company name, address, transaction date and total amount. These annotations enable the development of models that can jointly understand textual content and document layout to perform structured information extraction.

SROIE2019 is widely used as a benchmark for evaluating models designed for document layout understanding and key information extraction, particularly models that combine textual and spatial information such as LayoutLM-based architectures.

2.2 Dataset Structure and Labels

The dataset consists of two main components: receipt images and corresponding annotations.

Each sample in the dataset includes:

- Receipt image: a scanned or photographed receipt in image format.
- OCR annotations: text segments detected on the receipt along with their bounding box coordinates.
- Key information labels: ground truth values for the main entities that need to be extracted.

The primary information fields annotated in the dataset include:

- Company – the name of the store or vendor issuing the receipt
- Address – the store address
- Date – the transaction date
- Total – the total payment amount

The annotations provide both textual content and spatial information, which allows models to learn relationships between text and layout.

2.3 Dataset Characteristics

SROIE2019 exhibits several characteristics that make it suitable for training document understanding models.

First, the dataset contains receipts with diverse layouts, reflecting real-world variations across different vendors. The position and formatting of key information fields may vary significantly between receipts.

Second, the dataset provides structured annotations for key-value extraction, enabling supervised training for information extraction tasks.

Third, the dataset is considered relatively clean compared to many real-world document datasets, meaning that most receipts are clearly scanned and the annotations are well aligned with the text regions. Because of this, the data preparation process mainly focuses on formatting and normalization rather than extensive data cleaning.

Finally, the dataset supports research on multimodal document models that integrate text, spatial layout and visual information, making it particularly suitable for training models such as LayoutLMv3.

3. Data Preparation

3.1 Data Preparation Pipeline

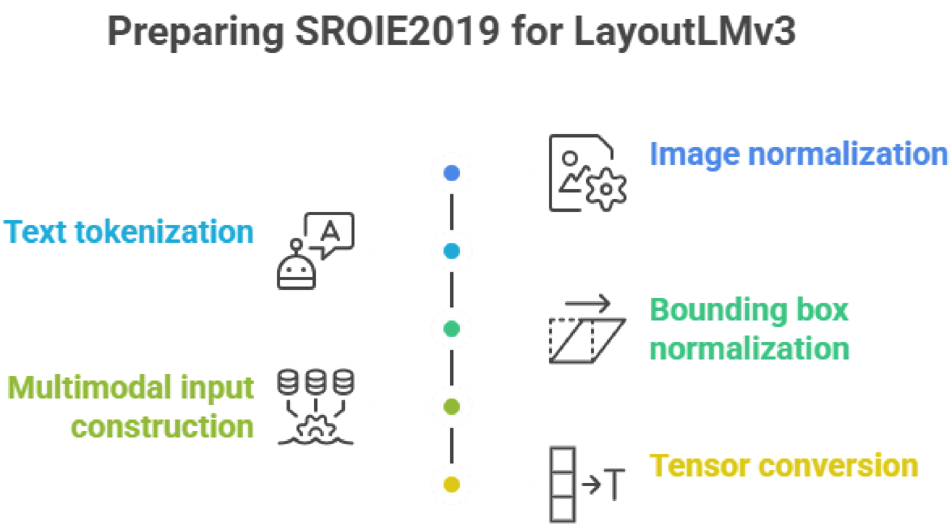

Figure 3.1 Data Preparation Pipeline

Before training the model, the SROIE2019 dataset must be transformed into a format compatible with the LayoutLMv3 architecture. The dataset provides receipt images along with OCR annotations containing textual content and bounding box coordinates. Since the dataset is relatively clean, the preprocessing stage mainly focuses on normalization and formatting rather than extensive data cleaning.

The data preparation pipeline consists of several steps including image normalization, text tokenization, bounding box normalization, multimodal input construction and tensor conversion. These steps ensure that the textual, spatial and visual modalities satisfy the input requirements of LayoutLMv3.

3.2 Image Normalization

Receipt images in the dataset may have different resolutions. To ensure consistent input for the model, each image is resized to a fixed resolution of 224×224, which corresponds to the input size expected by the visual encoder of LayoutLMv3.

Let I denote the original image and I' denote the resized image. The resizing operation can be expressed as:

$$I' = Resize(I, 224, 224)$$

After resizing, pixel values are normalized using mean and standard deviation normalization:

$$I_{norm} = \frac{I' - \mu}{\sigma}$$

where μ and σ denote the mean and standard deviation of pixel intensities. The final normalized image tensor can be represented as:

$$I_{norm} \in R^{3\times224\times224}$$

where the three channels correspond to the RGB color space.

3.3 Text and Bounding Box Processing

Each receipt image is associated with OCR annotations containing text segments and their corresponding bounding box coordinates. These annotations must be transformed into a format suitable for the LayoutLMv3 model. The preprocessing includes text tokenization and bounding box normalization.

3.3.1 Text tokenization

The OCR annotations provide text at the word level. However, LayoutLMv3 utilizes the WordPiece tokenizer, which decomposes words into smaller subword tokens to handle rare or unknown words.

Let the sequence of words extracted from OCR be defined as:

$$W = \{w_1, w_2, \ldots, w_n\}$$

Each word is processed by the WordPiece tokenizer:

$$WordPiece(w_1) \rightarrow \{t_{i1}, t_{i2}, ..., t_{ik}\}$$

The resulting token sequence becomes:

$$T = \{t_1, t_2, ..., t_m\}$$

Each token is then mapped to its corresponding vocabulary index:

$$t_x = ID(t_i)$$

LayoutLMv3 restricts the maximum input sequence length to 512 tokens, therefore: $m \leq 512$

If the token sequence exceeds this limit, it is truncated. Special tokens [CLS] and [SEP] are added to mark the beginning and end of the sequence.

3.3.2 Bounding box normalization

Each OCR word has a bounding box defined by pixel coordinates:

$$b_i = (x_{min}, y_{min}, x_{max}, y_{max})$$

Since receipt images may vary in size, the coordinates must be normalized to a fixed scale. LayoutLM models use a coordinate range from 0 to 1000.

Given the original image width *W* and height *H*, the normalized coordinates are computed as:

$$x' = \frac{x}{W} \times 1000, y' = \frac{y}{H} \times 1000$$

The normalized bounding box is therefore represented as:

$$b'_i = (x'_{min}, y'_{min}, x'_{max}, y'_{max})$$

where $0 \leq x', y' \leq 1000$

When a word is split into multiple subword tokens during tokenization, the same bounding box is assigned to all tokens belonging to that word.

3.4 Input Representation for LayoutLMv3

LayoutLMv3 is a multimodal model that jointly processes textual, spatial and visual information. The input representation therefore consists of three components: image features, text tokens and bounding box coordinates.

The textual input is represented as a sequence of token IDs:

$$X = [x_1, x_2, ..., x_n], \ n \leq 512$$

The spatial layout information is represented as bounding box coordinates for each token:

$$B_i = (x_{min}, y_{min}, x_{max}, y_{max})$$

The visual modality is represented by the normalized image tensor:

$$I \in R^{3\times224\times224}$$

These three modalities are jointly processed by the LayoutLMv3 architecture to capture the relationships between text content and document layout.

3.5 Tensor Construction for Model Input

After preprocessing, the processed inputs are converted into tensors that can be fed into the model during training.

The token sequence is represented as a tensor of token IDs: $X \in R^{n}$

The bounding box coordinates are represented as: $B \in R^{n\times4}$

where each row corresponds to a bounding box associated with a token.

The image input is represented as: $I \in R^{3\times224\times224}$

For the token classification task, each token is assigned a label representing the corresponding entity type: $Y = [y_1, y_2, ..., y_n]$

where tokens corresponding to padding positions are assigned a special label value of -100 to ensure that they do not contribute to the training loss.

The final input to the model can therefore be represented as a tuple (X, B, I, Y) which contains textual tokens, spatial layout information, visual features and the corresponding labels used for model training.

4. Model Architecture

4.1 Overview of LayoutLMv3

LayoutLMv3 is a multimodal transformer model designed for document understanding tasks. The model jointly learns representations from textual content, spatial layout information and visual features extracted from document images. By integrating these three modalities, LayoutLMv3 is able to capture both semantic and structural relationships within documents.

Given a document image and its associated OCR annotations, the model receives tokenized text, normalized bounding box coordinates and visual features from the image. These inputs are converted into embeddings and processed by a stack of transformer encoder layers to produce contextual representations for each token.

In this work, the pretrained LayoutLMv3-base model is fine-tuned for receipt information extraction using a token classification formulation.

4.2 Multimodal Inputs

LayoutLMv3 integrates textual, spatial and visual information through embedding layers before feeding them into the transformer encoder.

For the textual modality, each token is mapped to a dense vector using a token embedding matrix:

$$e_i = E(x_i)$$

where x_i is the token ID and E denotes the embedding lookup table.

Spatial layout information is incorporated through bounding box embeddings. Each token is associated with a normalized bounding box defined by: $b_i = (x_{min}, y_{min}, x_{max}, y_{max})$

These coordinates are projected into layout embeddings using learnable embedding functions.

The visual modality is obtained from the document image. The input image is divided into fixed-size patches following the Vision Transformer (ViT) approach. If the patch size is $P \times P$, the total number of patches is computed as: $N = \frac{224 \times 224}{P^2}$

Each patch is flattened and projected into a visual embedding vector.

The textual embeddings, layout embeddings and visual embeddings are then combined and processed by the transformer encoder. The encoder applies the multi-head self-attention mechanism to capture contextual dependencies across tokens and image regions:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where Q,K and V represent the query, key and value matrices.

4.3 Fine-tuning for Information Extraction

To perform receipt information extraction, LayoutLMv3 is fine-tuned as a token classification model. The contextual representation of each token produced by the transformer encoder is passed through a linear classification layer.

Let h_i denote the contextual representation of token i. The classification logits are computed as:

$$z_i = Wh_i + b$$

where W and b are learnable parameters.

The probability of assigning token iii to class ccc is obtained using the softmax function:

$$P(y_i = c|h_i) = \frac{exp(z_{ic})}{\sum_{j=1}^{C} exp(z_{ij})}$$

The model is trained using the cross-entropy loss:

$$L = -\frac{1}{n}\sum_{i=1}^{n}\sum_{c=1}^{C} y_{ic} log(\widehat{y}_{ij})$$

where y_{ic} represents the ground-truth label and $\widehat{y}_{ij}$ denotes the predicted probability.

The model parameters are optimized using the AdamW optimizer, which updates the parameters according to:

$$\theta_{t+1} = \theta_t - \eta\left(\frac{\widehat{m}_t}{\sqrt{\widehat{v}_t + \epsilon}} + \lambda\theta_t\right)$$

where η is the learning rate and λis the weight decay coefficient.

During training, the model is fine-tuned for 10 epochs with a learning rate of 3×10^{-5} and a batch size of 2 for both training and evaluation. Mixed precision training (FP16) is enabled when GPU acceleration is available to reduce memory usage and accelerate computation.

5. Information Extraction Pipeline

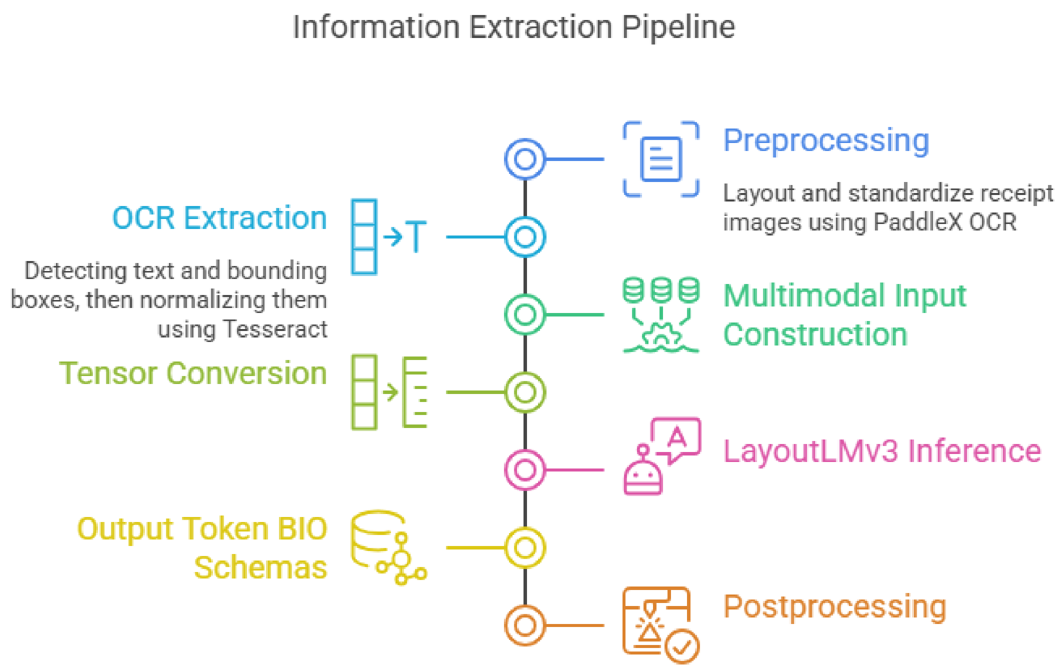

Figure 5 Information Extraction Pipeline

This section describes the pipeline used to extract structured information from receipt images. The process consists of three main stages: document normalization using PaddleX OCR preprocessing, text and bounding box extraction using Tesseract OCR and post-processing for entity aggregation. Fig. 1 illustrates the example receipt image used in this pipeline.

5.1 Input Receipt Image

The pipeline begins with a receipt image collected from real-world sources such as mobile camera captures, scanned documents, or digital receipts. These images may vary significantly in quality due to lighting conditions, background noise, or perspective distortion.

Before further processing, the system converts the input image into a consistent format suitable for document analysis. The objective of this stage is to preserve the visual structure of the receipt while preparing it for layout alignment and text detection.

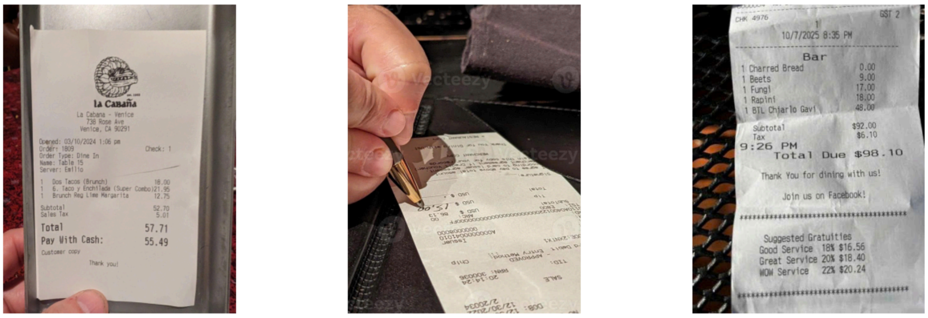

Figure 5.1 Examples of raw receipt images

5.2 Image Alignment and Image Normalization using PaddleX OCR

Receipts collected from real-world environments often suffer from geometric distortions such as skew, rotation, and perspective variations. These issues can negatively impact the performance of downstream OCR and document understanding models.

To address this, PaddleX OCR is employed as a preprocessing step for layout alignment and image normalization. The goal of this stage is to standardize the visual structure of receipt images before text extraction.

The alignment and normalization process consists of the following steps:

- Detecting text regions and estimating document orientation
- Applying geometric transformations to correct skewed or rotated images
- Standardizing the layout to improve spatial consistency

Figure 5.2 Example of raw receipt and corrected aligned receipt

After alignment, all receipt images are resized to a fixed resolution of 224 × 224 pixels, which matches the input requirements of the visual encoder in LayoutLMv3. This ensures that visual features are consistently represented across all samples.

By performing both layout alignment and image normalization at this stage, the system produces visually standardized inputs that are suitable for multimodal processing.

5.3 Text and Bounding Box Detection using Tesseract OCR

After image alignment and normalization, Tesseract OCR is applied to extract textual content and spatial information from the receipt images.

This stage includes three main components:

- **Text detection and recognition:** Tesseract identifies text regions within the image and extracts the corresponding textual content. Each detected text segment is associated with its bounding box coordinates.

574,766,637,766,637,817,574,817,La

676,759,876,759,876,817,676,817,Cabana

916,787,939,787,939,793,916,793,-

979,762,1183,762,1183,814,979,814,Venice

…

Figure 5.3 An example receipt image where text regions & OCR output

Figure 5.3, left, each frame represents a detected text region within the document. These frames indicate the location of text elements such as restaurant name, address, order details, item prices and total amount, allowing the system to grasp the spatial structure of the document, which is crucial for later stages of document comprehension and information extraction.

On the right side of figure 5.3, each detected text element is represented by a sequence of coordinates describing the four corners of the frame, followed by the recognized text content. In this format, numerical values represent the x and y coordinates of the frame corners, while the final element represents the recognized text within that region.

- **Text tokenization:** The extracted text is further processed using the WordPiece tokenizer. Each word is decomposed into subword tokens to ensure compatibility with the pretrained LayoutLMv3 model. The resulting token sequence is constrained to a maximum length of 512 tokens.
- **Bounding box normalization:** The original bounding box coordinates are defined in pixel space. To make them compatible with LayoutLMv3, the coordinates are scaled into a fixed range from 0 to 1000.

As a result, this stage produces structured multimodal inputs, including tokenized text sequences and normalized bounding boxes, which are directly fed into the LayoutLMv3 model.

5.4 LayoutLMv3 Inference

After normalization, the processed inputs are fed into the **LayoutLMv3** model for document understanding.

LayoutLMv3 is a multimodal transformer architecture designed for document AI tasks. Unlike traditional text-based models, LayoutLMv3 integrates three types of information simultaneously:

- textual tokens
- spatial layout information from bounding boxes
- visual features extracted from the document image

By jointly modeling these modalities, the model learns both semantic meaning and spatial relationships between text elements within the document.

During inference, the model performs token-level classification, assigning entity labels to each token. These labels correspond to predefined fields in receipt documents, such as merchant name, transaction date, total amount or product items.

5.5 Post-processing and JSON Output

The predictions produced by LayoutLMv3 are token-level labels that must be aggregated into meaningful structured fields. Therefore, a post-processing step is applied to convert the model outputs into structured information.

The post-processing stage includes the following steps:

1. Token merging: Consecutive tokens belonging to the same entity label are merged to form complete text segments.

2. Field aggregation: Extracted entities are grouped into predefined receipt attributes such as merchant name, date, tax and total amount.

3. Data structuring: The extracted information is converted into a structured JSON format, which can be easily used by downstream applications.

An example of the final output is shown below:


```
  "company": "",
  "address": "se SH la Canana La Cabana - Venice 738 Rose Ave Venice, CA 90291 1 1 1
1",
  "date": "03/10/2024 ",
  "total": "57.71"
```

The results show that the model successfully extracted the date and total amount from the receipt. The extracted values, *03/10/2024* and *57.71*, match the ground truth information in the document, indicating that the model performs well in identifying structured numerical fields.

However, the company entity was not detected, and the address field contains redundant and noisy tokens. Instead of extracting only the address information, the output includes additional text fragments such as duplicated numbers and incorrectly recognized tokens.

Several factors may contribute to this issue. First, OCR recognition errors may introduce noise into the extracted text, which can negatively affect the downstream entity extraction process. For example, the company name "La Cabana" appears as "la Canana" in the OCR output. Second, the proximity of textual elements in the receipt layout may cause the model to incorrectly group nearby tokens into the same entity. Finally, insufficient training samples for certain entity types may reduce the model's ability to distinguish between company names and address fields.

Overall, while the system demonstrates reliable performance in extracting structured numerical information, further improvements are needed to enhance the accuracy of text-based entities such as company names and addresses.

6. Evaluation

6.1 Evaluation Metrics

To evaluate the performance of the information extraction system, three standard metrics were used: Precision, Recall and F1-score. These metrics are widely used in Named Entity Recognition (NER) and information extraction tasks to measure the accuracy of predicted entities.

The evaluation metrics are defined as follows:

- Precision measures the proportion of predicted entities that are correct.

$$Precision = \frac{TP}{TP + FP}$$

- Recall measures the proportion of ground truth entities that are successfully detected.

$$Recall = \frac{TP}{TP + FN}$$

- F1-score is the harmonic mean of Precision and Recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

- Accuracy measures the proportion of correct predictions among all predictions made by the model.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

These metrics provide a comprehensive evaluation of the model’s ability to correctly detect and classify entities in document images.

6.2 Overall Performance

Model	Precision	Recall	F1-score	Accuracy
LayoutLM	0.67	0.52	0.56	0.52
LayoutLMv3	0.90	0.91	0.91	0.97

Table 6.2 The performance comparison LayoutLM and LayoutLMv3

The results demonstrate that LayoutLMv3 significantly outperforms the original LayoutLM model across all evaluation metrics.

LayoutLMv3 achieved a precision of 0.90, recall of 0.91 and F1-score of 0.91, indicating that the model is capable of accurately identifying and extracting key entities from invoice documents. Furthermore, LayoutLMv3 obtained an accuracy of 0.97, showing strong overall classification performance.

In contrast, the original LayoutLM model achieved lower performance, with a precision of 0.67, recall of 0.52 and F1-score of 0.56. The accuracy of 0.52 indicates that the model struggled to correctly classify entities within the document layout.

The superior performance of LayoutLMv3 can be attributed to its improved architecture that integrates textual, visual and layout information more effectively. By leveraging multimodal representations, LayoutLMv3 can better understand the spatial relationships and visual structure of documents, which is crucial for accurate information extraction.

These results confirm that LayoutLMv3 is more suitable for document understanding and invoice information extraction tasks compared to the earlier LayoutLM model.

6.3 Performance per Label

To further evaluate the model’s performance, the F1-score was analyzed for each entity label extracted from the invoices. The evaluated entity labels include ADDRESS, COMPANY, DATE and TOTAL.

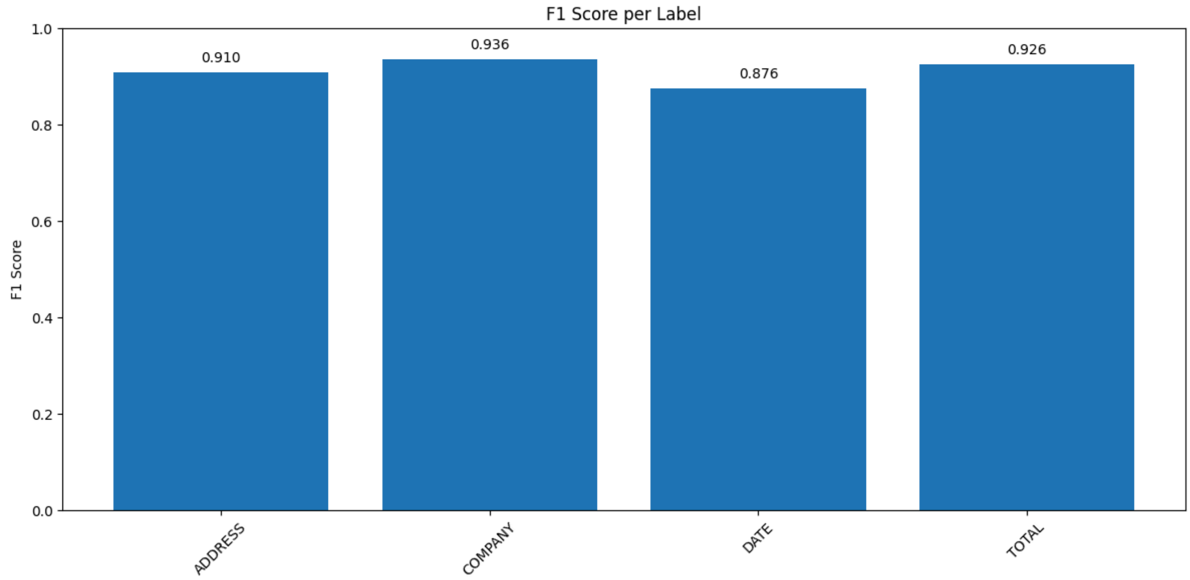

Figure 6.3 F1-score for each entity label

The results indicate that the model performs consistently well across all entity types, with F1-scores above 0.87.

- COMPANY achieved the highest performance with an F1-score of 0.936, indicating that company names are easier for the model to recognize due to their distinctive textual patterns.
- TOTAL achieved an F1-score of 0.926, demonstrating strong performance in identifying the total amount in invoices.
- ADDRESS obtained an F1-score of 0.910, showing that the model can effectively detect address fields despite variations in formatting and structure.
- DATE achieved the lowest score among the labels with an F1-score of 0.876, although the performance remains strong overall.

The relatively lower performance for DATE may be attributed to the variability in date formats across different invoices, such as different ordering of day, monthand year or variations in separators.

Overall, the results show that the proposed system provides highly accurate extraction across different entity types, confirming the robustness of the model for invoice information extraction tasks.

7. Error Analysis

Although the model achieves good performance overall, some prediction errors still occur in certain cases. One common issue arises when receipts do not contain specific fields such as the company name or address, making it difficult for the model to identify these entities. In addition, some receipts contain noisy text or background interference, which may lead to incorrect OCR outputs and affect the subsequent entity prediction.

Another challenge is the variation in receipt formats. In some cases, the positions of labeled fields are not fixed and may appear in different locations across receipts. This inconsistency can reduce the model's ability to accurately associate text tokens with their corresponding entities. Furthermore, low-quality input images, such as blurred or poorly scanned receipts, can also negatively impact OCR accuracy and overall extraction performance.

8. Conclusion

This study presented a receipt information extraction system based on the LayoutLMv3 model. The experimental results on the SROIE2019 dataset demonstrate that the proposed approach can effectively extract key information from receipt documents, achieving high performance in terms of precision, recall and F1-score.

However, several limitations were identified during the evaluation process. The extraction performance is influenced by errors in the OCR stage, particularly when processing low-quality receipt images. In addition, certain fields such as address and total amount remain challenging due to their variable formats and the presence of multiple numerical values in receipts.

In future work, the system can be improved by enhancing the image preprocessing stage before feeding the images into the model. Techniques such as noise reduction, contrast enhancement and image alignment can improve OCR quality and reduce recognition errors. Improving the quality of input images is expected to further enhance the accuracy and robustness of the receipt information extraction system.

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